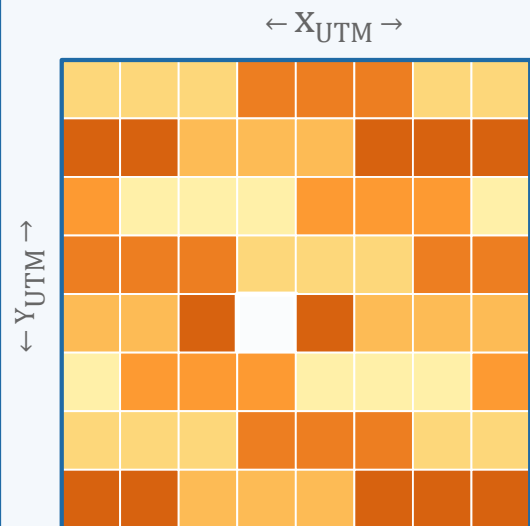
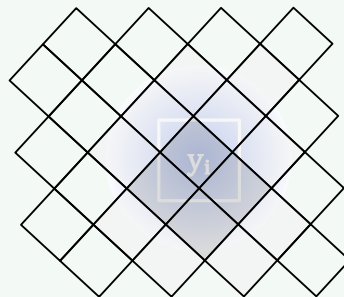


Input observation $y \in \mathbb{R}^N$



Dual-normalized
operators A, B



◇ HEALPix cell-average x_j

■ UTM pixel y_i

$$g_{ij} = \exp\left(-\frac{2d_{ij}^2}{s_{PSF}^2}\right) \quad j \in \mathcal{N}(i),$$

$$A_{ij} = \frac{g_{ij}}{\sum_{k \in \mathcal{N}(i)} g_{ik}}, B_{ji} = \frac{g_{ij} a_i}{\sum_{\ell: j \in \mathcal{N}(\ell)} g_{\ell j} a_\ell}$$

A: HEALPix→UTM

B: UTM→HEALPix

Damped residual
correction

$$x_{\text{ref}} = B y$$

$$(BA + \lambda I) \delta = B (y - A x_{\text{ref}})$$

$$\hat{x} = x_{\text{ref}} + \delta$$

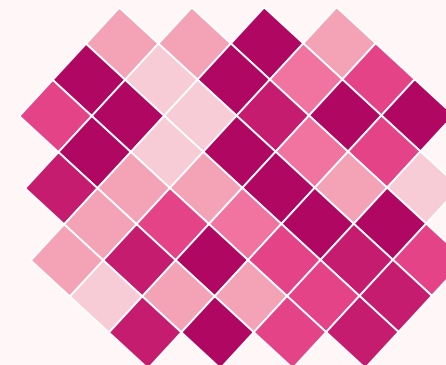
*sparse transfer
operators with a matrix-
free composite solve*

$O(qN)$ per iteration

no exact A^T and B^T required

**damping acts on
correction δ**

HEALPix field $\hat{x} \in \mathbb{R}^M$



Equal-area cells ·
Level = 20 · ≈ 6.3 m
stored as cell ids + values

optional $\hat{y} = A \hat{x}$ (round-trip consistency)

